

D228 Product Specification

Rev. 1.0 | Customer Review Version

D228 is an industrial LTE/NR gateway module designed for OEM equipment, industrial connectivity, remote device access, serial data applications, and compact embedded router deployments. The module provides cellular network access, dual Gigabit Ethernet ports, three serial ports (RS485/RS232/TTL), 5.8 GHz Wi-Fi, configurable I/O, optional GPS/BeiDou positioning, SMS management, VPN functions, cloud management, ZTP zero-touch provisioning, and support for remote management platforms. The cellular module model and supported bands can be confirmed according to the target region, carrier, and project configuration.

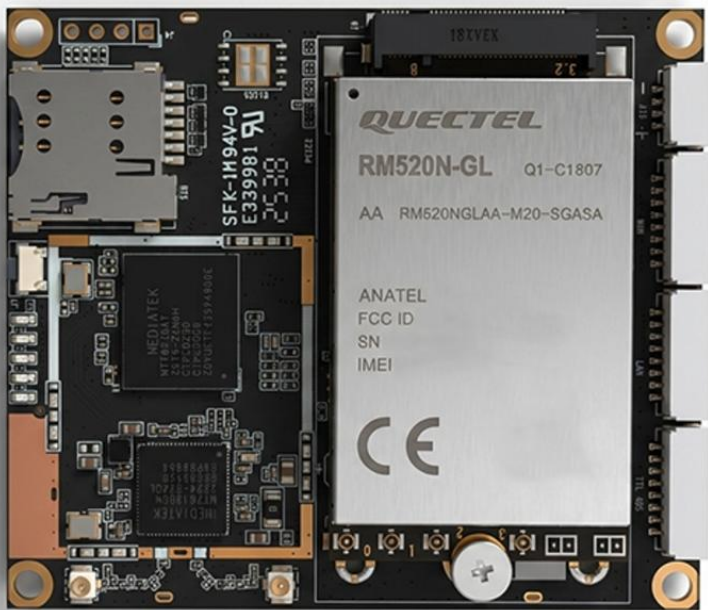


Table 0-1. Document Information

Document Field	Content
Product	D228 Industrial LTE/NR Router Module
Document Revision	Rev. 1.0

Document Status	Customer Review Version
Confidentiality	Customer Confidential
Scope	Engineering review, OEM integration, sample evaluation, and project communication
Document Date	2026-06-17

1. Product Overview

D228 is a board-level industrial gateway module designed for projects that need cellular access, Ethernet, Wi-Fi, serial ports, I/O, and remote management capabilities integrated into customer equipment. The module can operate as a compact industrial gateway or be expanded through the pin header base to expose additional interface resources, making it well-suited for OEM equipment, industrial field devices, and remote service systems.

The module is intended for embedded integration scenarios. Final enclosure design, antenna placement, power supply method, external interface breakout, and mounting structure should be reviewed and confirmed according to the customer project.

1.1 Key Capabilities

- Supports 3G/4G/5G cellular connectivity. Supported bands, carrier compatibility, and radio behavior depend on the selected cellular module, firmware version, and project configuration.
- Dual Gigabit Ethernet ports, with WAN/LAN roles changing based on operating mode.
- Three serial ports supporting RS485, RS232, and TTL.
- 5.8 GHz 802.11 a/an/ac Wi-Fi, suitable for hotspot, client, bridge, repeater, and multi-SSID scenarios.
- Two I/O channels with input/output switching support, and optional I2C configuration based on project requirements.
- Pin-header base expansion for additional Ethernet, serial, SIM, PCIe, USB 2.0, TF card, reset, LED, and I/O access.
- Optional eSIM support for improved deployment stability in place of a physical SIM card.
- Optional NAND Flash support for high-capacity storage and custom development environments.

- Supports GPS/BeiDou positioning, SMS management, standard VPN services, remote access, private networking, cloud management, and ZTP zero-touch provisioning.
- Supports serial passthrough, Modbus, MQTT, HTTP POST, and custom protocol conversion.
- Supports camera video overlay, allowing network status, GPS data, and serial or I2C data to be embedded into video streams.
- Built for industrial environments with wide-temperature support, hardware watchdog, ESD protection, and independent multi-rail power monitoring.
- Built on the Talonbox SkinOS embedded system platform, with support for component-based development, software package deployment, and project-level customization.

1.2 Target Applications

Table 1-1. Typical D228 Application Scenarios

Application	Typical Use
Industrial Gateway	Connect PLCs, sensors, camera, controllers, or field devices through cellular, Ethernet, Wi-Fi, serial, or IO paths
OEM Embedded Router	Integrate cellular routing and remote access capability into a customer enclosure or equipment platform
Remote Monitoring	Report device status, serial data, IO status, positioning data, or camera overlay information to a remote server or platform
Mobile And Field Equipment	Provide cellular access, Wi-Fi hotspot, VPN connectivity and remote real-time diagnostics for distributed assets
Custom Protocol Gateway	Implement customer-specific serial, IO, or remote management protocols through SDK and component development

1.3 System Platform

D228 uses the Talonbox SkinOS embedded system platform. SkinOS is a gateway operating system built by Talonbox on embedded Linux for modular development, remote management, protocol conversion, software package deployment, and field project customization. Project customers may request development documentation, SDK support, management tools, OTA tools, or custom component development according to project scope.

2. Appearance and Mechanical Information

2.1 Module Views

The following images show the D228 module form factor for product identification, mechanical review, and installation space planning.

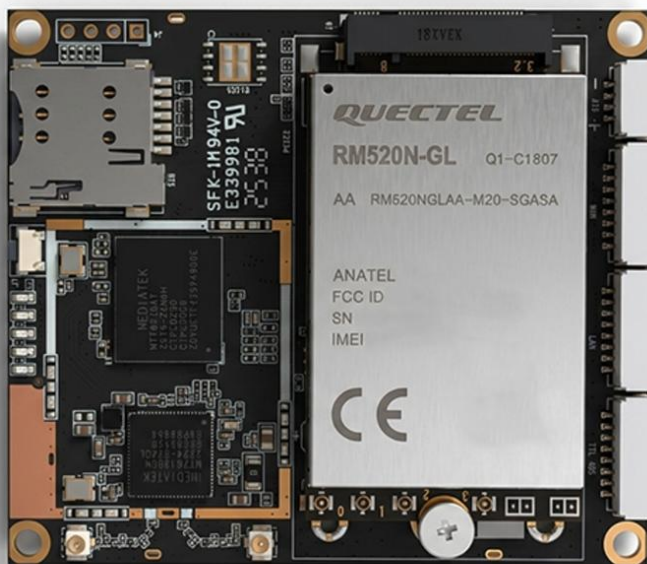


Figure 2-1. D228 front module view

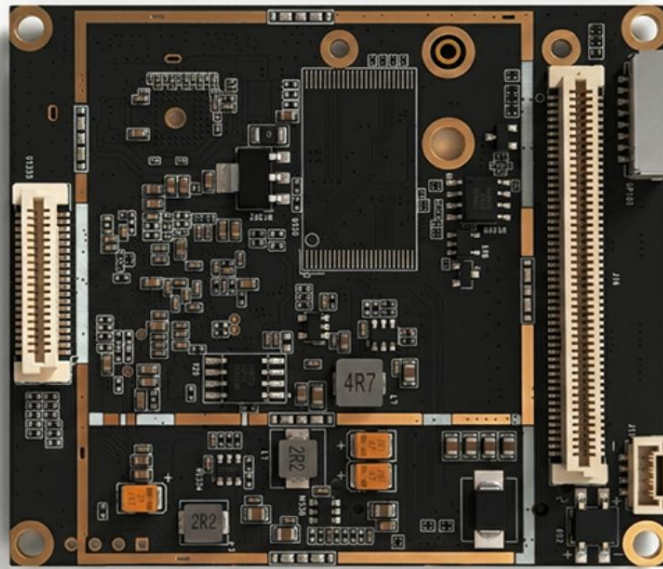


Figure 2-2. D228 rear module view



Figure 2-3. D228 side module view



Figure 2-4. D228 angled module view

2.2 Dimensions and Mounting

Table 2-1. D228 Mechanical Dimensions and Mounting Information

Item	Specification / Description
Overall Dimensions	Approx. 72.0 mm × 61.7 mm × 15.0 mm; The 15.0 mm value is the reference maximum module height / mechanical clearance height; The actual height shall be confirmed based on the final BOM, connector configuration, and sample measurement
Mounting Ear Hole Center Spacing	Approx. 67.1 mm × 56.7 mm, referring to the horizontal and vertical center-to-center spacing of the mounting ear holes; See the mechanical drawings for exact hole locations
Mounting Hole Description	A total of 4 locating/mounting holes are provided: Located on both sides of the module; Final hole diameter, hole spacing, keep-out/clearance area, and mounting screw specifications shall be confirmed based on the DXF file, mechanical drawings, and final mechanical design files
Main Interface Positions	The positions of the power connector, WAN/LAN expansion connectors, serial port, SIM card holder, reset button, LED SYS/Poor/Strong indicators, UART/IO expansion connectors, Wi-Fi/LTE antenna connectors, and other interfaces are shown in the front and rear mechanical drawings
Height And Mechanical Clearance	The 15.0 mm value is the reference maximum module height / mechanical clearance height; Mechanical design should reserve sufficient space for maximum component height, installed connector height, connector mating, antenna connector clearance, SIM

	card operation, cable routing, heat dissipation, and assembly tolerance
Mechanical Design Note	Enclosure openings, cable exit direction, interface clearance, standoff positions, connector clearance, antenna routing, heat dissipation conditions, and maintenance space should be confirmed based on the front and rear mechanical drawings, DXF file, and final sample verification

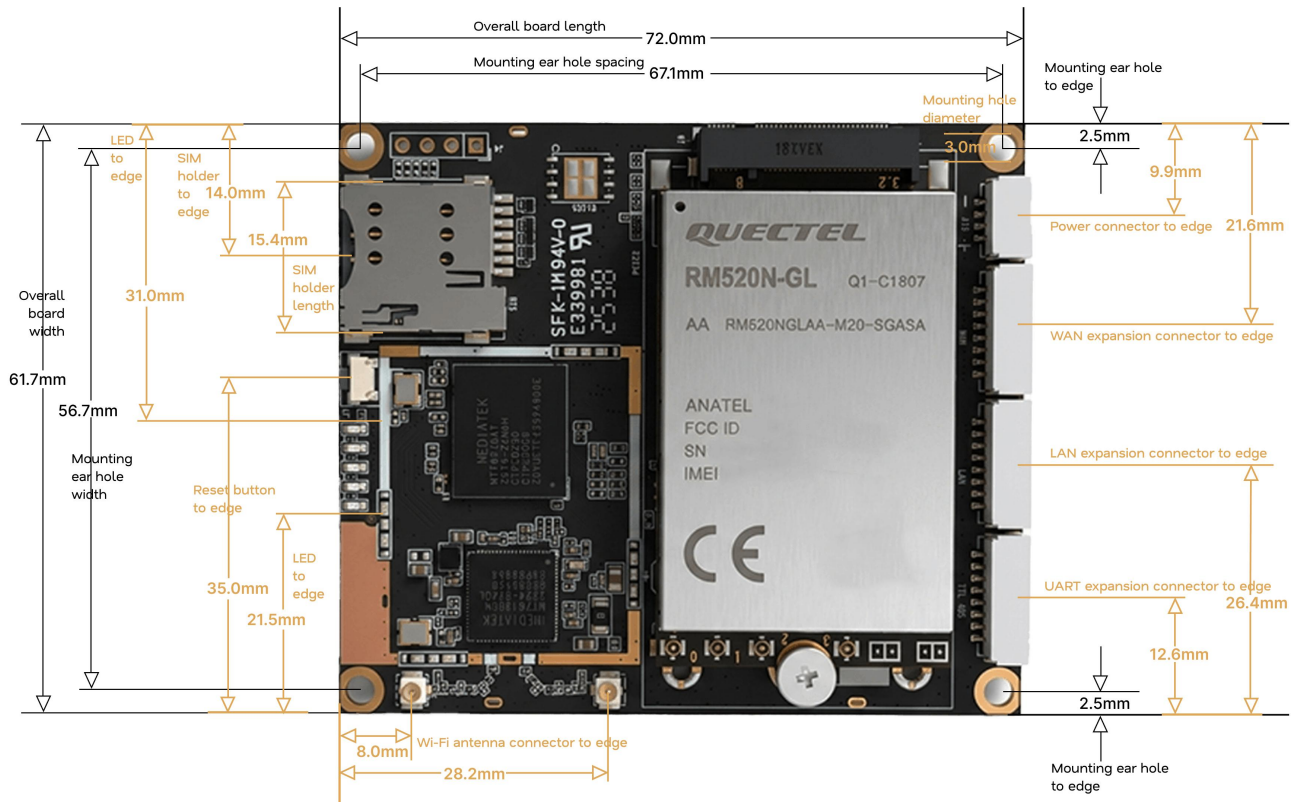


Figure 2-5. D228 Front Mechanical Dimension Drawing

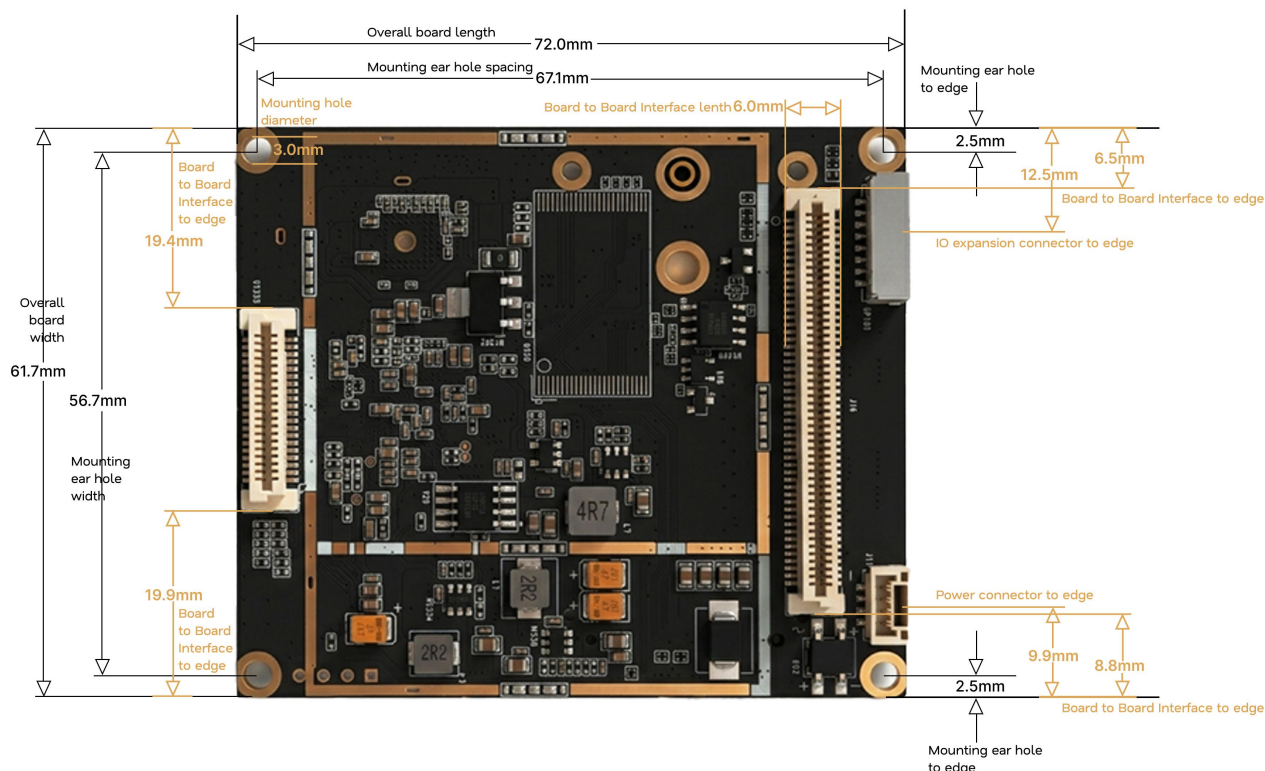


Figure 2-6. D228 Rear Mechanical Dimension Drawing

2.3 Handling and ESD Notes

- D228 is supplied as a PCBA module. Avoid direct contact with exposed components and connector pins during handling.
- Use an ESD-safe workstation, grounded wrist strap, and appropriate ESD packaging during assembly, inspection, and field replacement.
- Do not insert or remove the SIM card while the module is powered unless the project design explicitly supports this operation.
- Review final enclosure design, antenna routing, grounding, thermal design, and mounting method before deployment.

3. Hardware Specifications

The main D228 hardware specifications are listed below. Specific configurations may vary by cellular module, regional certification, hardware version, and project requirements.

Table 3-1. D228 Hardware Specification Summary

Parameter	Specification / Description
Main Processor	MediaTek MT7621DAT
Cellular Module	Standard reference configuration may use the Quectel RM520N-GL 5G module. The final module model may vary based on target region, certification, carrier, band support, and supply requirements
Cellular Connectivity	Supports 3G/4G/5G cellular connectivity depending on the installed module, firmware support, carrier requirements, and regional certification scope. Supported radio technologies depend on the selected module and project configuration
Ethernet	Two onboard 1000M Gigabit Ethernet ports: WAN and LAN. Port roles depend on the operating mode
Wi-Fi	5.8 GHz 802.11a/an/ac, 2T2R, with a theoretical PHY rate up to 866 Mbps. Reference Wi-Fi chipset is MediaTek MT7613BEN. Supports AP, client, bridge, repeater, and related firmware modes
Serial Ports	Three serial ports: TTL, RS232, and RS485, exposed through the external UART interface. Pinout and multiplexing are subject to the project interface definition
I/O	Two 3.3 V I/O channels with configurable input/output modes; selected I/O can be configured for I2C use, and the base can further expand additional I/O resources
DC Input	Two 4-Pin, 1.25 mm power connector positions, one horizontal and one vertical. Unless otherwise specified by the project, either power input position may be used
Input Voltage	7-15 V, recommended supply 12 V / 2 A; 5 V input support must be confirmed by project configuration
Power Consumption	Full-load power consumption < 5 W; instantaneous peak power may vary with radio environment, network state, and service load; reserve at least 5 W power budget
SIM Slot	Micro SIM slot; insert SIM with notch facing outward and metal contacts facing the PCB; Do not insert or remove the SIM while the device is powered on
Antenna	2 Wi-Fi IPEX-1 connectors; 4 LTE/5G IPEX-4 connectors; optional GPS IPEX-1 connector depending on the model/project configuration
Button	1 reset button
LEDs	5 status LEDs: LAN, WAN, Strong, Poor, and SYS, used for Ethernet link, cellular signal strength, and system operating status. LED behavior depends on firmware version and project configuration
Expansion Connectors	Rear dual-row board-to-board connector, 1.0 mm pitch, 20-pin + 50-pin (40 + 100 contacts total). Expands Ethernet, serial, SIM, PCIe, USB 2.0, TF, reset, LEDs, I/O, and power

Operating Temperature	Supports -40 °C cold start and long-term operation up to +70 °C
Protection And Reliability	Supports hardware watchdog, independent power monitoring, fault recovery, and scheduled/idle reboot mechanisms. Exact behavior depends on firmware version and project configuration

3.1 Cellular Communication Reference

The cellular parameters below use the Quectel RM520N-GL 5G module as a reference. Module selection, frequency bands, certification, antenna configuration, product labels, and carrier acceptance requirements for different target regions should be confirmed during the project phase.

Table 3-2. Cellular Communication Reference Parameters

Item	Reference Specification
Reference LTE/NR Module	Quectel RM520N-GL 5G module
5G bands	NR NSA: n41 / n78 / n79; NR SA: n1 / n28* / n41 / n77 / n78 / n79
4G bands	FDD LTE: B1 / B2 / B3 / B5 / B7 / B8 / B20 / B28; TDD LTE: B34 / B38 / B39 / B40 / B41
3G bands	WCDMA: B1 / B2 / B5 / B8
MIMO	DL 4x4: n1 / n41 / n78 / n79; DL 2x2: n28; UL 2x2: n41 / n77 / n78 / n79; LTE DL: 2x2
Bandwidth	SA Sub-6: DL 2 Gbps / UL 1 Gbps; 5G NSA Sub-6: DL 2.2 Gbps / UL 575 Mbps; LTE: DL 600 Mbps / UL 150 Mbps; UMTS: DL 42.2 Mbps / UL 11 Mbps
Output Power	Subject to selected cellular module datasheet

3.2 Wi-Fi Communication Reference

Table 3-3. Wi-Fi Communication Reference Parameters

Item	Reference Specification
Standard	IEEE 802.11a/an/ac, 5.8 GHz
MIMO	2T2R
Theoretical PHY Rate	Up to 866 Mbps
Transmit Power Reference	5G HT20 MCS7: 15 dBm; 5G HT40 MCS7: 15 dBm; 5G AC40 MCS9: 13 dBm; 5G AC80 MCS9: 13 dBm

Receiver Sensitivity Reference	5G HT20 MCS7: -65 dBm; 5G HT40 MCS7: -62 dBm; 5G AC40 MCS9: -54 dBm; 5G AC80 MCS9: -51 dBm
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4. Interface Resources and Electrical Definitions

This section describes the main D228 interface locations, electrical definitions, and expansion interface resources. The annotated interface images are provided for quick identification of components and connector positions. Specific signal definitions, wiring methods, and project configuration requirements are defined in the corresponding tables.

4.1 Front Interface Layout

The front side of D228 integrates the cellular module, SIM slot, reset button, WAN/LAN, serial interfaces, power input, and antenna interfaces. Figure 4-1 is provided for quick identification of main front-side interfaces and component positions. Figure 4-2 provides detailed pin arrangement and signal definitions for each connector.

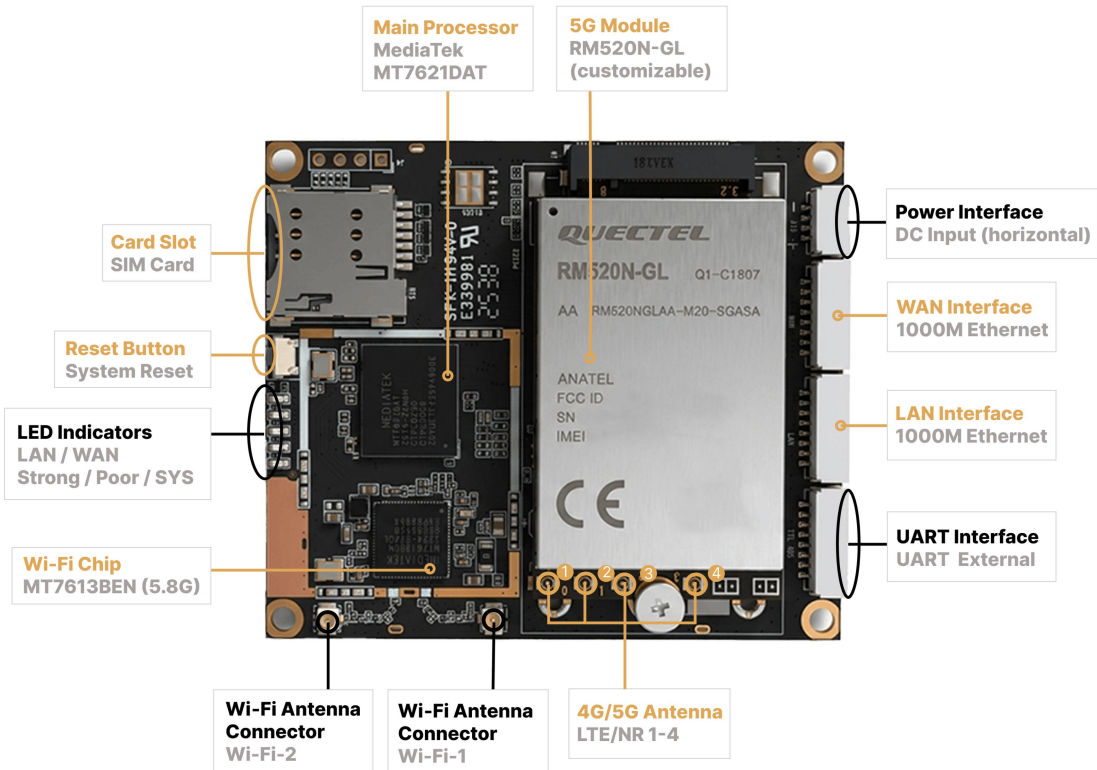


Figure 4-1. D228 front interface and major component layout

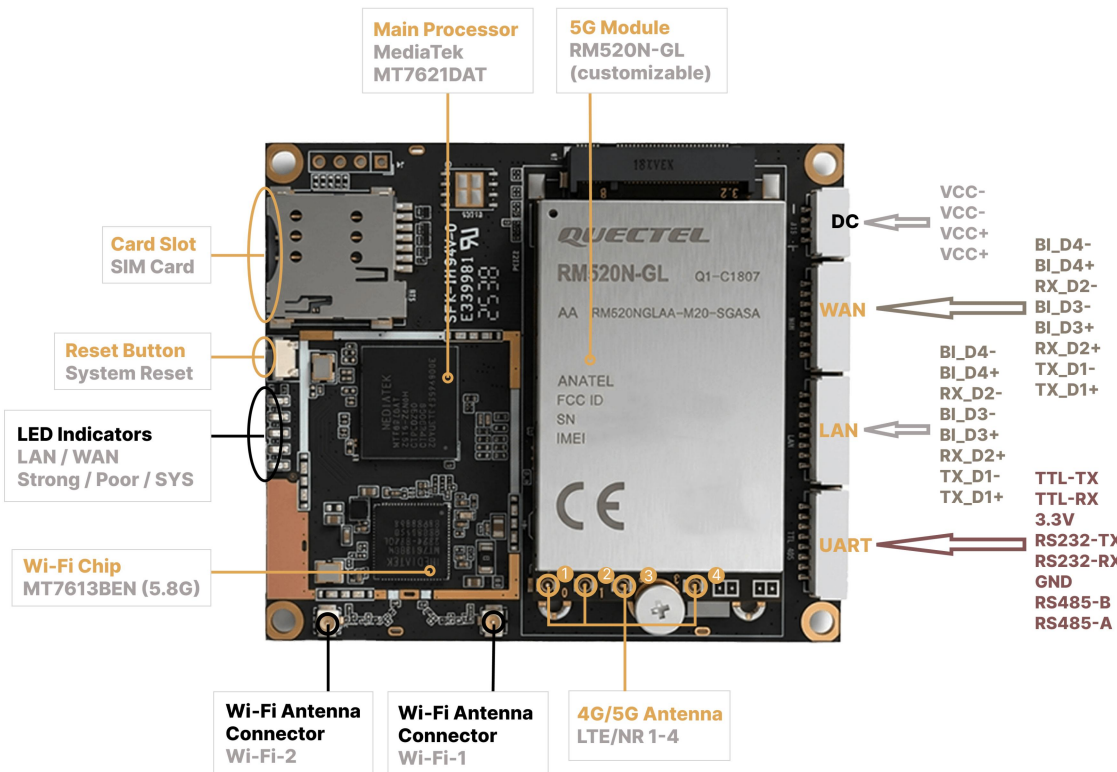


Figure 4-2. D228 front interface pin marking

Table 4-1. Front Interface Definitions

Interface	Definition / Description
Horizontal DC Input	4-Pin, 1.25 mm power connector on the front side. The module provides both horizontal and vertical DC input positions. Unless otherwise specified by the project, either power input may be used
SIM Slot	Micro SIM slot; insert with notch facing outward and contacts facing the PCB
Reset Button	After system startup, press and hold for about 5-8 seconds and release to restore factory settings and reboot
LAN / WAN / Strong / Poor / SYS LED Group	Five front-side status indicators for LAN link, WAN link, strong cellular signal, poor cellular signal, and system operation status
WAN Interface	1000M Gigabit Ethernet interface. Used as WAN in gateway or hybrid WAN modes; may operate as LAN in other modes
LAN Interface	1000M Gigabit Ethernet interface for local wired device access

UART External	External serial interface for TTL, RS232, RS485, and related serial resources. Pinout and function assignment are subject to the project interface definition
LTE/NR Antennas	Four 4G/5G antenna connectors, marked LTE/NR 1-4, for cellular main/diversity and MIMO antenna connections
Wi-Fi Antennas	Two 5.8 GHz Wi-Fi antenna connectors, marked Wi-Fi-1 and Wi-Fi-2

Table 4-2. Front Connector Pin Definitions

Connector	Pin	Signal	Description
DC Input	1	VCC+	Power positive
DC Input	2	VCC+	Power positive
DC Input	3	VCC-	Power negative
DC Input	4	VCC-	Power negative
WAN	1	TX_D1+	Gigabit Ethernet transmit differential positive
WAN	2	TX_D1-	Gigabit Ethernet transmit differential negative
WAN	3	RX_D2+	Gigabit Ethernet receive differential positive
WAN	4	BI_D3+	Gigabit Ethernet bidirectional differential
WAN	5	BI_D3-	Gigabit Ethernet bidirectional differential
WAN	6	RX_D2-	Gigabit Ethernet receive differential negative
WAN	7	BI_D4+	Gigabit Ethernet bidirectional differential
WAN	8	BI_D4-	Gigabit Ethernet bidirectional differential
LAN	1	TX_D1+	Gigabit Ethernet transmit differential positive
LAN	2	TX_D1-	Gigabit Ethernet transmit differential negative
LAN	3	RX_D2+	Gigabit Ethernet receive differential positive
LAN	4	BI_D3+	Gigabit Ethernet bidirectional differential
LAN	5	BI_D3-	Gigabit Ethernet bidirectional differential
LAN	6	RX_D2-	Gigabit Ethernet receive differential negative

LAN	7	BI_D4+	Gigabit Ethernet bidirectional differential
LAN	8	BI_D4-	Gigabit Ethernet bidirectional differential
UART	1	RS485-A	RS485 serial A
UART	2	RS485-B	RS485 serial B
UART	3	GND	Ground
UART	4	RS232-RX	RS232 serial receive
UART	5	RS232-TX	RS232 serial transmit
UART	6	3.3V	3.3V power output
UART	7	TTL-RX	TTL serial receive
UART	8	TTL-TX	TTL serial transmit

4.1.1 WAN/LAN Wiring Reference

D228 WAN/LAN ports use Gigabit Ethernet differential signals. For harness or adapter board design, refer to the pin map below. The final design should be confirmed with the Ethernet magnetics, ESD protection, cable length, and mechanical constraints of the project.

Table 4-3. WAN/LAN Wiring Reference

Interface Signal	RJ45 Reference Pin	Description
TX_D1+ / WAN-TX+ / LAN-TX+	1	Transmit differential positive signal
TX_D1- / WAN-TX- / LAN-TX-	2	Transmit differential negative signal
RX_D2+ / WAN-RX+ / LAN-RX+	3	Receive differential positive signal
BI_D3+	4	Bi-directional differential pair
BI_D3-	5	Bi-directional differential pair
RX_D2- / WAN-RX- / LAN-RX-	6	Receive differential negative signal
BI_D4+	7	Bi-directional differential pair
BI_D4-	8	Bi-directional differential pair

Auto-negotiation Note	-	Ethernet supports auto-negotiation; routing should still follow standard pinout and impedance requirements
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4.2 Rear Interface Layout

The rear side of D228 includes the 20-Pin / 50-Pin base connectors, LED/GPIO/serial expansion interfaces, power input, and Gigabit Ethernet-related interfaces. The diagram is used to quickly identify the main rear interfaces and functional areas.

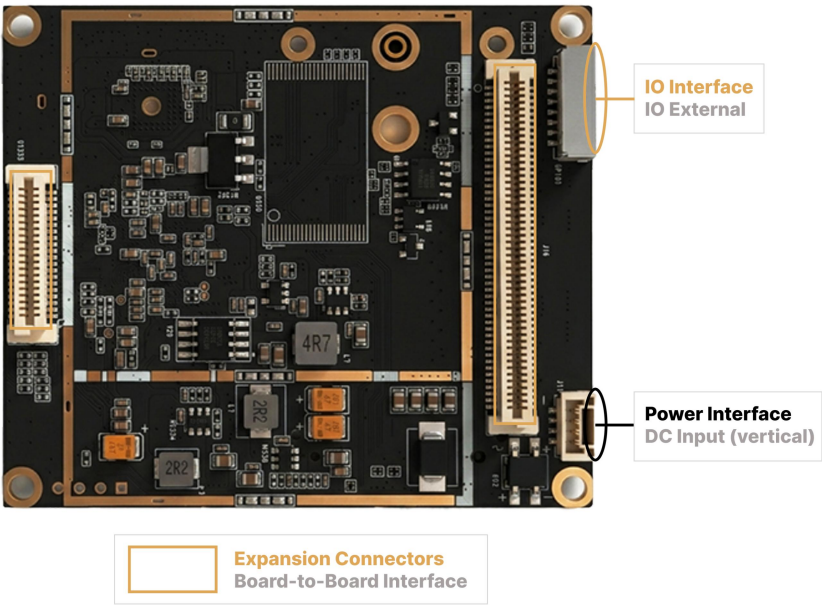


Figure 4-3. D228 rear interface and functional area layout

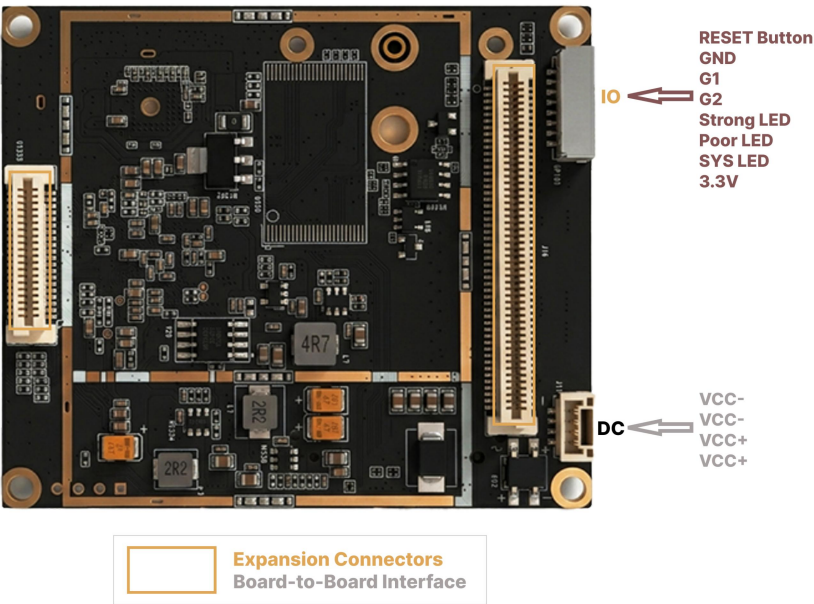


Figure 4-4. D228 rear interface pin marking

Table 4-4. Rear Interface Definitions

Interface	Definition / Description
Vertical DC Input	4-Pin, 1.25 mm power connector on the rear side. The module provides both horizontal and vertical DC input positions. Unless otherwise specified by the project, either power input may be used
IO External	External I/O interface for I/O, LED, reset, 3.3V, GND, and related signals. It can be configured for input/output control or I2C expansion depending on project requirements
Board-to-Board Expansion Connectors	Rear-side board-to-board expansion connectors for Ethernet, serial, SIM, PCIe, USB 2.0, TF card, reset, LED, I/O, and power interface expansion
Expansion Connector Area	Intended for OEM carrier boards, adapter boards, or customer system integration. Pin assignment, power design, and interface multiplexing should be confirmed during project hardware review

Table 4-5. Rear Connector Pin Definitions

Connector	Pin	Signal	Description
IO	1	3.3V	3.3V power output
IO	2	SYS LED	System indicator pin

IO	3	Poor LED	Poor signal indicator pin
IO	4	Strong LED	Strong signal indicator pin
IO	5	G2	I/O 2, configurable as I2C SDA
IO	6	G1	I/O 1, configurable as I2C SCL
IO	7	GND	Ground
IO	8	RESET Button	System reset button
DC Input	1	VCC-	Power negative
DC Input	2	VCC-	Power negative
DC Input	3	VCC+	Power positive
DC Input	4	VCC+	Power positive

4.2.1 Indicator Status Description

LED states can be used for installation, commissioning, and troubleshooting. Details may vary by firmware version, and the final LED behavior is subject to the project firmware.

Table 4-6. LED Status Description

LED	Status	Meaning
SYS LED	Solid after power-up	The module is powered on and the system is in the early boot stage
SYS LED	Slow blinking during boot/self-check	The system is booting or performing self-check. If abnormal slow blinking persists, check the cellular module, SIM card, and firmware status
SYS LED	Fast blinking during dialing or upgrade	The module is dialing the cellular network or performing an upgrade operation
SYS LED	Solid on after network connection	The system has completed booting and successfully connected to the network
Poor Signal LED	Solid	Cellular signal is weak
Poor Signal LED	Blinking	Cellular signal is very weak. Check antenna connection, installation position, and carrier coverage

Strong Signal LED	Blinking	Cellular signal is good
Strong Signal LED	Solid	Cellular signal is strong
WAN LED	Solid	A wired device is connected to the WAN port
WAN LED	Blinking	Data is being transmitted through the WAN port
WAN LED	Off	No active WAN connection is detected
LAN LED	Solid	A wired device is connected to the LAN port
LAN LED	Blinking	Data is being transmitted through the LAN port
LAN LED	Off	No active LAN connection is detected

4.3 20-Pin / 50-pin Connector Expansion

D228 provides rear-side 20-Pin / 50-Pin board-to-board expansion connectors for extending Ethernet, serial, USB 2.0, PCIe, SIM-related connections, reset, LED, I/O, I2C, NAND Flash, and power interfaces. Pin group functions should be reviewed together with the project carrier board design, power architecture, interface multiplexing, and firmware configuration.

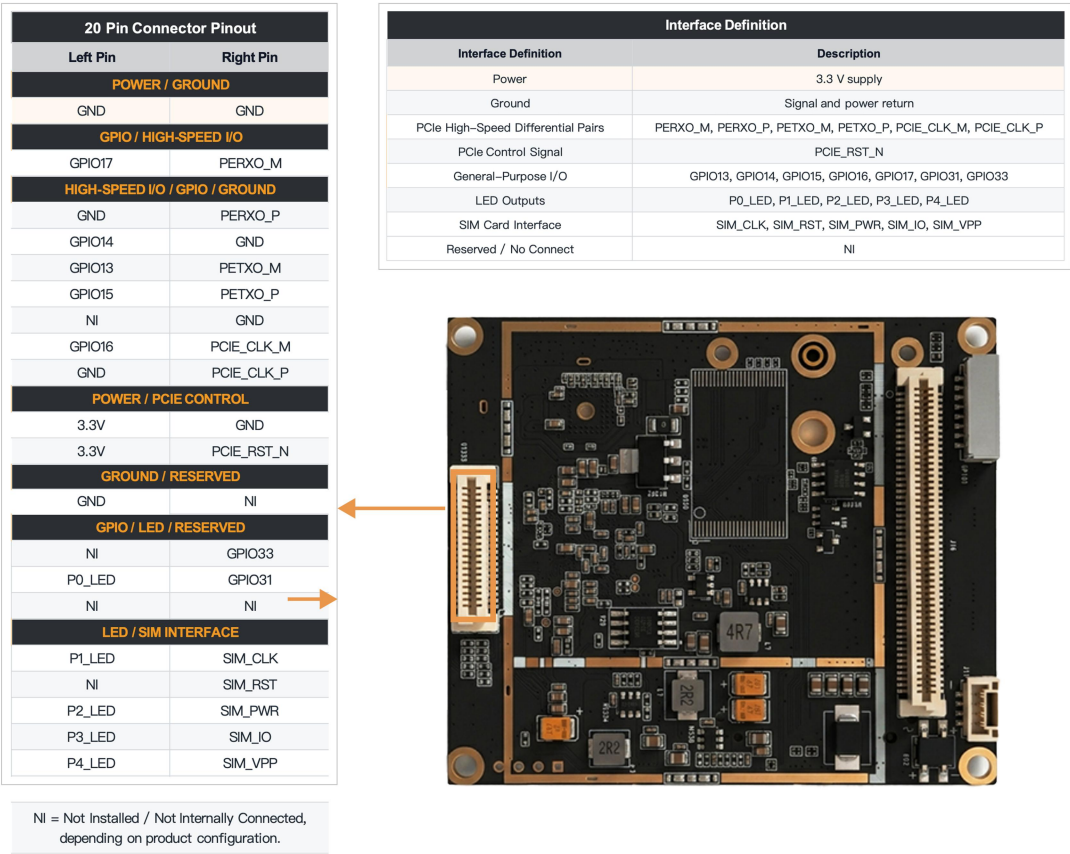


Figure 4-5. D228 rear 20-Pin Expansion Connector Layout, Pinout, and Interface Definition

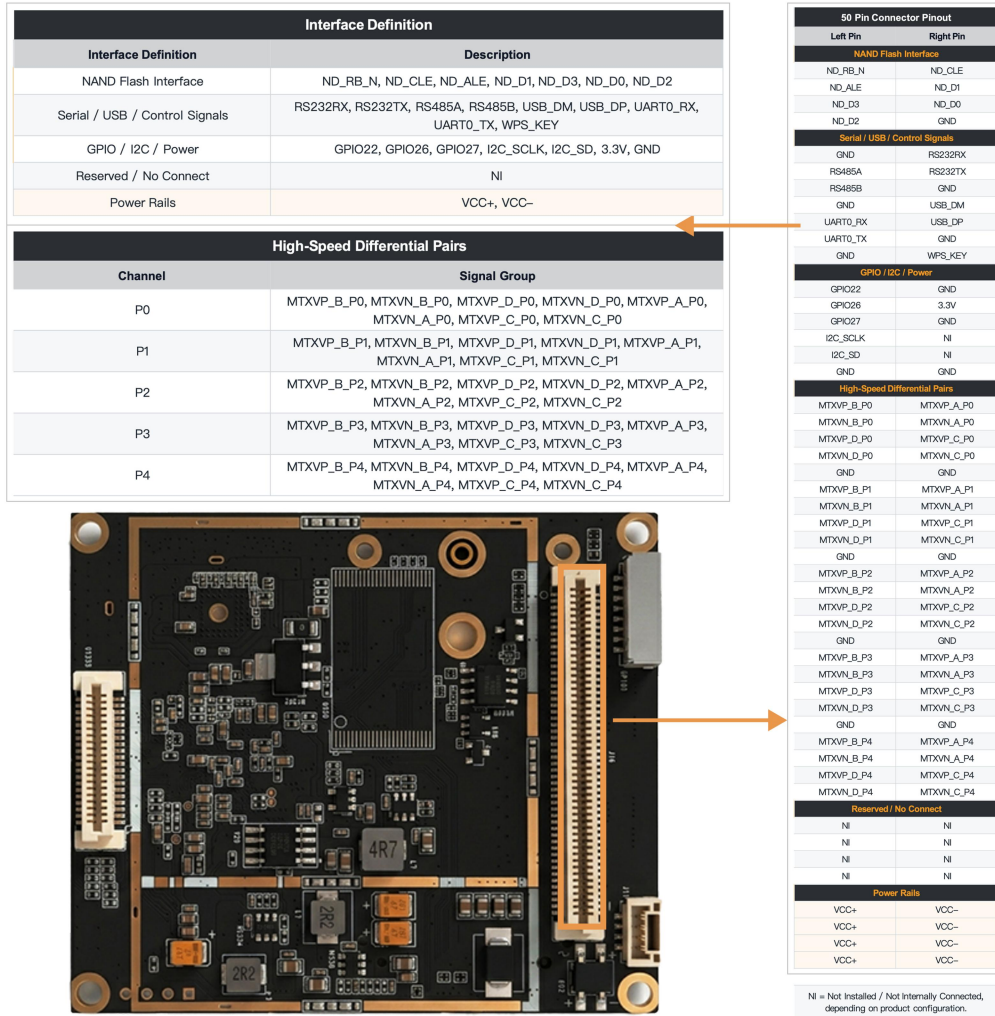


Figure 4-6. D228 rear 50-Pin Expansion Connector Layout, Pinout, and Interface Definition

Table 4-7. 20-Pin / 50-Pin Expansion Connector Pin Group Function Description

Pin Group	Function / Description
VCC+ / VCC-	Power and ground pins, supporting 7-15V input. When powering through the base connector, multiple VCC+ and VCC- pins should be used in parallel to improve power reliability and current capacity
RS485A / RS485B	RS485 serial signals, shared with the onboard UART RS485 interface. Simultaneous use is not recommended
RS232RX / RS232TX	RS232 serial signals, shared with the onboard UART RS232 interface. Simultaneous use is not recommended
UART0_TX / UART0_RX	TTL serial signals, shared with the onboard UART TTL interface. Simultaneous use is not recommended

USB_DM / USB_DP	USB 2.0 differential signals for external USB 2.0 expansion
I2C_SCLK / I2C_SD	I2C interface signals; may also be configured as I/O resources depending on project requirements
GPIO22 / GPIO26 / GPIO27	GPIO signals associated with SYS, Poor Signal, and Strong Signal LEDs. Final behavior depends on project firmware and hardware definition
WPS_KEY	Reset key signal corresponding to the system RESET function
MTXVP/MTXVN_P0	Gigabit Ethernet LAN1 differential signals. External RJ45 design requires Ethernet magnetics and appropriate ESD protection
MTXVP/MTXVN_P1	Gigabit Ethernet LAN2 differential signals. External RJ45 design requires Ethernet magnetics and appropriate ESD protection
MTXVP/MTXVN_P2	Gigabit Ethernet LAN3 differential signals. External RJ45 design requires Ethernet magnetics and appropriate ESD protection
MTXVP/MTXVN_P3	Gigabit Ethernet LAN4 differential signals, shared with the onboard LAN interface
MTXVP/MTXVN_P4	Gigabit Ethernet WAN differential signals, shared with the onboard WAN interface
ND_RB_N / ND_CLE / ND_ALE / ND_D0-D3	NAND Flash interface signals. External NAND Flash design should follow the Talonbox reference design and matching software support
GPIO17 / GPIO14 / GPIO13 / GPIO15 / GPIO16 / GPIO33 / GPIO32	Extended I/O signals for project-specific customization
PERX0_M/P / PETX0_M/P / PCIE_CLK_M/P / PCIE_RST_N	PCIe interface signals for external PCIe wireless cards or other PCIe devices; software support is required
P0_LED-P4_LED	Ethernet Link LED signals for LAN1-LAN4 and WAN

During mechanical design and project introduction, confirm the 20-Pin / 50-Pin expansion connector model, pad layout, body height, mating direction, and mating connector.

Table 4-8. 20-Pin / 50-Pin Expansion Connector Mechanical and Project Confirmation Items

Item	Description
Connector Type	Rear-side 20-Pin / 50-Pin board-to-board expansion connectors; exact part number is subject to the final BOM and project configuration

Supported Functions	Ethernet, serial, USB 2.0, PCIe, SIM-related connections, reset, LED, I/O, I2C, NAND Flash, and power expansion
Mechanical Review	Before mechanical design, confirm connector pitch, pad layout, body height, mating direction, mating connector, and assembly space
Power Usage	When powering through the base connector, multiple VCC+ and VCC- pins should be used in parallel to improve power stability
Notes	Final dimensions, pads, and mechanical mating requirements are subject to project-confirmed connector specification, BOM, and mechanical files

4.4 Onboard Connector Selection and Mechanical Review

Table 4-9. Onboard Connector Selection and Mechanical Review

Item	Description
Connector Type	1.25 mm pitch horizontal/vertical board connectors and rear-side board-to-board expansion connectors; exact part numbers are subject to the final BOM and project configuration
Applicable Interfaces	WAN, LAN, UART, I/O, power, antenna, SIM, reset, and expansion connector interfaces
Mechanical Review	Before mechanical release, confirm connector pitch, mating terminals, body height, mating direction, assembly space, and harness exit direction
Supply Confirmation	Before mass production, confirm connector model, mating terminals, lead time, and lifecycle supply status

5. Connectivity and Routing Capabilities

5.1 Internet Access Modes

- Supports 3G/4G/5G cellular connectivity, with Modem/passthrough mode available depending on the cellular module, firmware, and project configuration.
- Supports wired WAN access through configurable Gigabit Ethernet WAN behavior.
- Supports upstream Wi-Fi access through 5.8 GHz Wi-Fi client, bridge, or repeater modes.
- Supports hybrid multi-WAN operation using combinations of cellular, wired WAN, and Wi-Fi paths.
- Supports configurable failover, hot standby, load balancing, and link priority based on operating mode and project policy.

- Supports project-specific intelligent access policies, such as concurrent multi-link operation, client- or application-based path selection, and dynamic access switching based on field network conditions.

5.2 Cellular Network Functions

- Supports multiple dialing methods, including point-to-point dialing and network-card-style dialing.
- Automatically matches carrier parameters based on the carrier configuration database.
- Supports custom APN, SIM PIN, SMS management, RAT lock, band lock, cell lock, SIM lock, module lock, and custom AT query/configuration.
- IPv6 protocol stack support is available depending on firmware and project configuration.

5.3 Network Reliability

- Supports multiple network keepalive mechanisms, including ICMP detection and received-packet count detection.
- Supports configurable recovery thresholds and fault recovery policies to help keep field devices online.
- May support hardware watchdog, independent power unit monitoring, low-temperature self-heating, and scheduled/idle reboot functions, subject to firmware version and project configuration.
- Supports HA/VRRP for multi-gateway redundancy and high-availability deployment scenarios.

6. Service Protocols and Application Capabilities

6.1 Serial Protocol Capabilities

- Supports serial transparent transmission with configurable frame length, frame interval, flow control, custom registration packet, keepalive packet, custom prefix/suffix, custom activation packet, multiple servers/clients, multi-protocol operation, and traffic statistics.
- Supports multi-data-center serial passthrough for remote data collection and multi-platform reporting scenarios.
- Supports Modbus RTU to Modbus TCP conversion for PLCs and industrial real-time devices.
- Supports MQTT serial data transparent transmission and server-issued activation packet behavior.
- Supports HTTP POST for reporting serial data to a specified server.

- Supports command mode based on proprietary HE commands for gateway control and management.
- Supports parsing and reporting external GPS/BeiDou data through serial interface.
- Supports custom serial protocol conversion through SDK and component development.

6.2 IO and Industrial Control

- Supports I/O input/output mode configuration for remote control, status collection, and industrial field automation.
- Supports MQTT-based IO management, including topic subscription and IO state-change event publishing.
- Supports custom remote IO management protocols for server-side real-time IO operation and status notification.
- After the gateway is connected to the remote platform, IO can be managed through platform APIs or the platform interface.
- I/O pins can be configured as I2C depending on project design, for external devices such as displays or battery gauges.

6.3 Positioning Capabilities

D228 can support GPS/BeiDou positioning data collection and multi-protocol reporting depending on project configuration.

Table 6-1. Positioning Capabilities and Data Reporting Methods

Function	Description
Positioning Data Source	Supports GPS/BeiDou positioning capability and may also support an external serial positioning module depending on project configuration
NMEA Reporting	Supports NMEA raw data reporting with configurable header and reporting interval
MQTT Reporting	Supports positioning data reporting through MQTT
HTTP Reporting	Supports positioning data reporting through HTTP JSON format
JT/T808	JT/T808 positioning data upload can be supported by project configuration; specific function depends on firmware version and platform integration requirements

Platform Access	After connecting to the remote management platform, positioning data can be viewed and managed through the platform
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6.4 Network Application Capabilities

Table 6-2. Network Application Function List

Capability Group	Supported Functions
Firewall And NAT	Firewall, port forwarding, DMZ host, port proxy, ALG, static routing, source-address routing, port routing
Dynamic Services	DDNS, UPnP, domain redirect, domain hijacking, IGMP forward/reverse proxy
Routing Protocols	RIPv1, RIPv2, RIPv6, OSPFv2, OSPFv3
Monitoring	Real-time network traffic graph, terminal traffic statistics, terminal traffic control, access control, scheduled access, WOL
Wireless Services	5.8 GHz AP, 5.8 GHz client, wireless bridge/repeater, whitelist/blacklist, transmit power adjustment, optional Portal/ad push customization
Time Management	Base-station time sync, NTP, LAN time service for downstream devices
Video Overlay	When supported by project integration, network signal, GPS information, and sensor data can be overlaid onto supported camera video streams

6.5 VPN and Private Networking

- PPTP client with MPPE support.
- L2TP client with tunnel password support.
- GRE tunnel.
- OpenVPN client with pre-shared key or certificate support.
- IPsec with pre-shared key or certificate support.
- WireGuard.
- Self-organizing networking, NAT traversal, and site-to-site networking can be supported depending on platform configuration, firmware version, and project design.

6.6 NAT Traversal and Cross-site Networking

D228 can support NAT traversal and cross-site networking by project configuration for remote access to field devices, multi-site networking, and remote maintenance scenarios. Specific functions, network scale, security policy, and platform dependencies should be confirmed during the project phase.

Table 6-3. NAT Traversal and Cross-site Networking Capabilities

Function	Description
NAT Traversal	Maps TCP/UDP services of downstream gateway devices to a remote access entry by project configuration for remote maintenance of field devices
Remote Access	Can be used to access gateway Web UI, downstream device management pages, serial services, or other project-specified services
Cross-site Networking	Supports multi-site networking, remote subnet interconnection, and secure device-to-device access by project configuration
Security And Permissions	Specific authentication method, access permissions, encryption policy, and platform configuration are subject to the project plan

7. Management, Maintenance, and Open Integration

7.1 Local and Remote Management

- Supports local TCP control protocol, local HTTP status query, and local broadcast discovery.
- Provides local Linux management toolkit and programming API for gateway management and OTA upgrade.
- Supports reboot, reset, status query, and configuration changes through SMS.
- Supports remote HTTP status reporting, including gateway information, terminal information, GPS/BeiDou positioning information, and other operating status.
- Supports remote HTTP control, MQTT status reporting, alarms, and MQTT-based gateway management.
- Remote management server SDK or interface support can be provided on a project basis, enabling customers to integrate cloud-based management, OTA updates, remote gateway management, API integration, and feature extensions in a Linux server environment.

7.2 Remote Operations Capabilities

D228 can work with the remote management platform to provide device status review, batch maintenance, remote login, configuration management, and related operations capabilities. Specific platform functions, permission scope, and data interfaces are subject to project configuration.

Table 7-1. Remote Operations Capability Description

Capability	Description
Batch Status Review	Can review gateway online status, network state, cellular signal, terminal information, traffic, operating status, and related data
Batch Maintenance	May support batch reboot, batch reset, batch upgrade, and maintenance task delivery according to platform capability
Remote Login	May support remote entry to gateway Web UI, remote terminal, SSH/Telnet, or serial CLI; specific access method depends on project configuration
Downstream Device Access	Can access management pages or service ports of downstream gateway devices by project configuration for remote debugging and maintenance
Configuration Backup/Delivery	May support gateway configuration backup, configuration delivery, and batch configuration management by project configuration
Platform Interface	Remote management API or SDK can be provided for customer platform integration and secondary development

7.3 Web UI and Quick Setup



Figure 7-1. Web UI login page

Table 7-2. Web UI Default Access Information

Item	Default / Description
Default Management IP	192.168.8.1 or project-defined management IP

Default Username	admin or project-defined account
Default Password	admin or project-defined credential; production projects should define secure credential handling
Default Wi-Fi SSID	Default SSID may use device model identifier and MAC address suffix; specific prefix can be configured by project
Default Wi-Fi Password	Project-defined; default credentials should be reviewed before production deployment
Factory Reset	After startup, press and hold RESET for about 5-8 seconds and release; the module restores factory settings and reboots where supported
Web UI Customization	Logo, language, default configuration, user permission model, and restricted operation policies can be customized by project

7.4 Common Operating Modes

Table 7-3. Common Operating Modes

Mode	Description
4G/5G Cellular Gateway	Accesses the Internet through 3G/4G/5G cellular networks and provides network connectivity to connected clients
Wired Gateway	Accesses the Internet through wired WAN and provides Internet access to downstream clients
Wi-Fi Client/Repeater Gateway	Connects to an upstream Wi-Fi network through 5.8 GHz Wi-Fi client, bridge, or repeater modes
Wireless Access Point	Operates as a 5.8 GHz wireless access point for downstream Wi-Fi clients
Hybrid Multi-WAN Mode	Combines cellular, wired WAN, and Wi-Fi access paths for cold standby, hot standby, or load balancing
Combined Access Mode	Supports project-specific combinations such as cellular + bridge, cellular + wired LAN, and cellular + wireless LAN
Modem/Passthrough Mode	Supports 3G/4G/5G passthrough mode depending on the cellular module, firmware, and project configuration

7.5 Upgrade, Customization, and Development Support

D228 is built on the Talonbox SkinOS embedded system platform and can provide firmware upgrade, configuration management, Web UI customization, SDK/API, component-based software package

deployment, and script development support according to project requirements. These capabilities support customer protocol adaptation, business logic extension, permission management, and batch deployment.

Table 7-4. Upgrade, Customization, and Development Support

Item	Description
Configuration Management	Supports configuration backup/import, default configuration customization, and retaining specified settings after factory reset where supported
Firmware Upgrade	Supports local upgrade, online upgrade, and LAN batch upgrade; x86 and aarch64/ARM gateway management toolkit and OTA upgrade tools can be provided by project
Web UI Customization	Supports logo upload, Web UI customization, language configuration, normal-user/super-user interface separation, and permission restriction policies
FPK Software Package	Supports component-based software package installation, uninstallation, recovery, and deployment for project function extension and custom feature delivery
Online Scripts	May support online script development, event-triggered execution, scheduled execution, and packaged deployment by project configuration
SDK/API Support	Local management API, remote management interface, development documentation, and customer platform integration support can be provided by project
Protocol Customization	Serial, IO, remote management, or custom protocol conversion logic can be customized by project requirement
Operations Restriction Policies	Configuration, upgrade, reboot, factory reset, and configuration import/export operations can be locked or restricted based on project configuration

8. Model Configurations

D228 is available in different configurations based on cellular module integration method, positioning capability, NAND Flash, interface expansion, and project customization requirements. The table below describes common configuration differences. Final model name, cellular module, interface configuration, and availability are subject to project confirmation.

Table 8-1. D228 Model Configuration Description

Model	Main Difference	Description
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D228 Standard	Integrated 4G/5G LTE/NR global cellular module	Suitable for typical industrial cellular gateway, OEM embedded router, and remote device access applications
D228-4G	Integrated 4G LTE full-band cellular module	Suitable for projects that only require 4G LTE connectivity
D228-GPS	Integrated 4G LTE global cellular module with GPS positioning support	Suitable for projects requiring positioning data collection and reporting
D228 Core	M.2 connector for customer-supplied 5G NR module	Module compatibility, antenna configuration, firmware support, thermal design, and certification scope should be confirmed before use
D228-4G Core	MiniPCle slot for customer-supplied 4G LTE module	Suitable for projects using a customer-specified LTE module or an already certified module
D228-NAND	Integrated 4G/5G LTE/NR global cellular module with NAND Flash support	Suitable for projects requiring additional storage, software package deployment, or a custom development environment

9. Ordering and Project Documentation Notes

Table 9-1. Ordering, Lead Time, and Project Documentation Notes

Item	Description
Evaluation Sample	Available by project request. Provide target region, carrier requirements, antenna plan, interface requirements, installation environment, and intended application
OEM Configuration	Firmware defaults, remote platform settings, branding, protocol behavior, interface definition, IO behavior, enclosure integration, and test requirements can be reviewed by project
Engineering Files	Mechanical drawings, interface layout, connector reference, pinout, electrical notes, DXF/STEP files, and engineering review package can be provided during project evaluation
Lead Time	Standard configurations in stock may ship within 7 business days as a reference lead time; final timing depends on quantity, cellular module option, certification requirements, component supply, testing plan, and customization scope
MOQ	Sample and production MOQ should be confirmed by project
Project Confirmation	Final configuration, frequency bands, antennas, labels, certification scope, software functions, and delivered materials are subject to project-confirmed version

10. Certification, Compliance, and Configuration Notice

- Certification marks shown on module-level components apply to the cellular module only. They do not automatically indicate final D228 board-level certification or end-product certification.
- The final customer product, including enclosure, antenna placement, power design, connector/interface design, installation method, labeling, and final integration, may require additional certification or engineering review.
- Cellular module selection, supported frequency bands, antenna configuration, product labels, reference images, carrier acceptance, and delivery timing may vary by target market and project configuration.
- Specifications may vary due to component supply, firmware version, hardware version, regional requirements, or project customization.

11. Contact and Project Review

To request an evaluation sample, OEM integration review, mechanical files, cellular module selection guidance, or remote management platform support, please contact Talonbox with your project requirements, including target region, carrier requirements, expected quantity, enclosure plan, installation environment, and required interfaces.

Recommended information to include

Target region, carrier requirements, expected quantity, enclosure plan, required interfaces, serial/IO requirements, and whether the project requires a standard D228 configuration or an OEM integration review.

- Email: sales@talonbox.com
- Website: <https://www.talonbox.com>
- Inquiry form: <https://www.talonbox.com/#contact>

Final configuration, delivery timing, certification scope, and engineering file availability are confirmed by project.